

PATTERN RECOGNITION USING SPIKING NEURAL NETWORKS

PROJECT REPORT

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**DEPARTMENT OF ELECTRONICS & COMMUNICATION
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DECLARATION

We, hereby declare that the project report “**PATTERN RECOGNITION USING SPIKING NEURAL NETWORKS**” submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology in Electronics and Communication Engineering of APJ Abdul Kalam Technological University, Kerala, is a bonafide work done by us under the supervision of Prof. Sajeena A (Associate Professor of the Dept of ECE, TKM College of Engineering). This submission represents our ideas in our own words, and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained.

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CERTIFICATE

This is to certify that the project report entitled **“PATTERN RECOGNITION USING SPIKING NEURAL NETWORKS”** is a bonafide record of the work presented by **AFTAB MUHAMMED FAISAL (TKM21EC012), ANAND P (TKM21EC027), ANANTHU C (TKM21EC030), ANIRUDH S (TKM21EC031)** to APJ Abdul Kalam Technological University in partial fulfillment of the requirement for the award of degree of Bachelor of Technology in Electronics and Communication Engineering during the academic year 2024-2025 under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

Spiking Neural Networks (SNNs) have emerged as a promising approach for energy-efficient neuromorphic computing, combining bio-inspired learning with event-driven processing. This paper presents a compact, fully CMOS-based implementation of an on-chip trainable SNN system featuring a 2×3 memristive crossbar array with integrated Spike-Timing-Dependent Plasticity (STDP) synapses and Leaky Integrate-and-Fire (LIF) neurons.

The proposed architecture implements a modified STDP learning rule that enables in-situ training without requiring external computational resources. A 2×3 crossbar configuration with CMOS-based memristive synapses demonstrates efficient pattern recognition of three distinct 2×1 pixel patterns. The system incorporates a Winner-Take-All (WTA) mechanism at the output layer for robust classification.

Implemented in 180nm CMOS technology, the design achieves low-power operation while maintaining resilience against process variations and noise. Post-layout simulation results validate the system's functionality, showing successful training and recognition of the target patterns. The compact 2×3 crossbar implementation serves as a scalable building block for larger neuromorphic systems while demonstrating the feasibility of fully integrated CMOS solutions for edge computing applications.

This work advances the development of practical neuromorphic hardware by providing a complete, transistor-level implementation of an SNN system with on-chip learning capabilities. The successful demonstration of this small-scale system paves the way for larger array implementations in future research.

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Abbreviations and Notations

- **ANN:** Artificial Neural Network
- **CMOS:** Complementary Metal-Oxide-Semiconductor
- **IC:** Integrated Circuit
- **LIF:** Leaky Integrate-and-Fire
- **MOSFET:** Metal-Oxide-Semiconductor Field-Effect Transistor
- **PSTDP:** Post-Synaptic Spike-Timing-Dependent Plasticity
- **PVT:** Process, Voltage, and Temperature
- **SNN:** Spiking Neural Network
- **STDP:** Spike-Timing-Dependent Plasticity
- **VLSI:** Very Large-Scale Integration
- **WTA:** Winner-Take-All

Chapter 1

INTRODUCTION

1.1 GENERAL BACKGROUND

The continuous evolution of Very Large-Scale Integration (VLSI) and neuromorphic computing has paved the way for advanced hardware implementations capable of mimicking biological neural processes. Traditional Artificial Neural Networks (ANNs) have shown remarkable success in various machine learning applications, yet they remain computationally expensive and power-intensive due to their reliance on Von Neumann architectures. This limitation has prompted researchers to explore bio-inspired models such as Spiking Neural Networks (SNNs), which offer an energy-efficient alternative by utilizing event-driven spike-based computations.

Among various neural computing models, SNNs are recognized for their ability to process information in a way that closely resembles the human brain. Unlike conventional ANNs, where neurons communicate through continuous activations, SNNs transmit information through discrete spikes, thereby significantly reducing power consumption and improving processing efficiency. The efficacy of SNNs is further enhanced through Spike-Timing-Dependent Plasticity (STDP), a biologically plausible learning mechanism that enables synaptic weights to adapt based on the temporal correlation of pre- and post-synaptic neuron activities. This synaptic plasticity mechanism is essential for on-chip learning and real-time pattern recognition tasks.

However, the implementation of SNNs in hardware presents several challenges, particularly in designing efficient and scalable synapses and neuron models. Traditional CMOS-based implementations of SNNs face limitations in terms of area efficiency and scalability. To address these concerns, memristive devices have been proposed as an alternative for synaptic weight storage and plasticity implementation. Memristors, with their non-volatile memory properties and analog-like behavior, enable compact and efficient hardware realization of STDP learning rules. This project focuses on the integration of CMOS-based memristive STDP synapses with Leaky Integrate-and-Fire (LIF) neurons to achieve an on-chip trainable SNN system capable of performing real-time pattern recognition.

The motivation behind this project is to develop a low-power, compact, and efficient neuromorphic hardware system that can be utilized for various edge computing applications. By leveraging CMOS technology and memristive synapses, this work aims to contribute to the advancement of neuromorphic computing and provide a viable solution for real-world intelligent processing systems.

1.2 ORGANISATION OF CHAPTERS

The upcoming chapters highlight the following contents:

1. Chapter 2 deals with the literature review, providing an overview of existing research and developments in the field of Spiking Neural Networks.
2. Chapter 3 briefs about Spiking Neural Networks, discussing their structure, functionality, and relevance in modern computing.
3. Chapter 4 discusses the Leaky Integrate and Fire neuron model, detailing its design and operational principles.
4. Chapter 5 deals with Spike Timing Dependent Plasticity (STDP), explaining its significance and implementation in SNNs.
5. Chapter 6 focuses on the crossbar array architecture implemented in this project, detailing its configuration and role in facilitating the connectivity between neu-

rons and memristive synapses.

6. Chapter 7 covers the working mechanism of the implemented system, explaining how the spiking neurons interact with the memristive synapses and the STDP learning rules applied for weight adjustments.
7. Chapter 8 discusses the different modes of operation of the system, including the training and inference modes, and how they contribute to the overall functionality of the spiking neural network.
8. Chapter 9 presents an overview of the Cadence Virtuoso simulation environment, highlighting its features, tools, and advantages for circuit design and analysis crucial to this project.
9. Chapter 10 analyzes the results obtained from the simulations conducted in Cadence, interpreting their significance in the context of neuromorphic computing and the performance of the designed circuits.
10. Chapter 11 concludes the report by summarizing the key findings, reflecting on the limitations encountered, and offering insights into future directions for research and potential applications of the developed systems.

Chapter 2

Literature Survey

2.1 Introduction

Recent advancements in neuromorphic computing have highlighted Spiking Neural Networks (SNNs) for their capability to mimic biological neural processing using spike-based communication. This literature review focuses on recent developments in SNN implementations, particularly in integrated circuit designs that enhance efficiency and performance.

Key studies address CMOS technology, innovative memory architectures, and learning mechanisms like spike-timing-dependent plasticity (STDP), all while striving for energy-efficient solutions. The review also covers encoding techniques that improve signal processing in SNNs. Overall, this chapter provides an overview of the current landscape in SNN research and its implications for future neuromorphic systems.

2.2 Overview of Spiking Neural Networks and Integrated Circuits

2.2.1 CMOS Circuit Implementation of Spiking Neural Network for Pattern Recognition

[1] S. K. Vohra et al., "CMOS Circuit Implementation of Spiking Neural Network for Pattern Recognition Using On-chip Unsupervised STDP Learning," in *Integration, The VLSI Journal*, vol. 95, 2024.

This paper presents a CMOS circuit implementation of a spiking neural network (SNN) specifically designed for pattern recognition tasks, leveraging on-chip unsupervised spike-timing-dependent plasticity (STDP) learning to enhance efficiency in recognition. The proposed design incorporates a compact layout that facilitates effective learning and recognition capabilities, allowing the SNN to adaptively adjust its connections based on the temporal patterns of input spikes. The authors emphasize the advantages of hardware-level implementations in providing efficient, low-power solutions suitable for real-time applications in robotics and artificial intelligence. Through detailed experimental validation, the study demonstrates the robustness and scalability of the circuit in recognizing various patterns, showcasing its potential impact on future neuromorphic systems.

2.2.2 Hardware Architecture for Columnar-Organized Memory

[2] J. Shamsi, K. Mohammadi, and S. B. Shokouhi, "A Hardware Architecture for Columnar-Organized Memory Based on CMOS Neuron and Memristor Crossbar Arrays," in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 26, no. 12, pp. 2355-2368, Dec. 2018.

The authors propose a novel hardware architecture for columnar-organized memory that integrates CMOS neurons with memristor crossbar arrays, creating a hybrid system that combines neural processing with memory storage. This design aims to enhance the efficiency and scalability of hardware implementations for neural networks by reducing the physical area and power consumption typically associated with tradi-

tional memory architectures. The study includes comprehensive simulation results that illustrate the architecture's capability to significantly lower energy usage while maintaining high performance. The authors discuss the implications of this architecture for large-scale neural network implementations, indicating its potential to support complex computational tasks in machine learning and cognitive computing.

2.2.3 PSTDP Circuit for Analog Implementation of SNNs

[3] I. Satyaraj and B. J. Kailath, "A Simple PSTDP Circuit for Analog Implementation of Spiking Neural Networks," in 2020 IEEE 4th Conference on Information & Communication Technology (CICT), pp. 1-6, 2020.

This paper introduces a straightforward and effective PSTDP (Post-Synaptic Spike-Timing-Dependent Plasticity) circuit tailored for the analog implementation of spiking neural networks. By closely mimicking biological learning processes, the proposed circuit simplifies the design complexity often associated with SNN hardware. The authors present experimental results that validate the circuit's performance in various configurations and environments, demonstrating its adaptability and effectiveness in capturing the essential dynamics of spiking neurons. The findings underscore the circuit's potential applications in real-time processing tasks, contributing to the development of more efficient neuromorphic systems that can operate with minimal energy consumption.

2.2.4 55nm CMOS Analog Circuit Implementation of LIF and STDP Functions

[4] Z. Yang et al., "55nm CMOS Analog Circuit Implementation of LIF and STDP Functions for Low-Power SNNs," in 2021 IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED), pp. 1-6, Jul. 2021..

The authors present a 55nm CMOS analog circuit that effectively implements leaky integrate-and-fire (LIF) neuron models alongside STDP functions, which are crucial for developing low-power spiking neural networks (SNNs). The work carefully examines the trade-offs between performance metrics—such as speed and accuracy—and power consumption, providing insights into the design methodologies that facilitate ef-

ficient analog processing. The authors employ rigorous simulation and experimental techniques to validate the circuit's functionality, emphasizing its suitability for neuromorphic applications that require energy-efficient solutions without sacrificing performance. This study contributes to the ongoing effort to create sustainable hardware for advanced neural computing.

2.2.5 Energy- and Area-Efficient CMOS Synapse and Neuron Circuits

[5] B. Joo et al., "Energy- and Area-Efficient CMOS Synapse and Neuron for Spiking Neural Networks With STDP Learning," in IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 69, no. 9, pp. 3632-3642, Sept. 2022.

This paper discusses the design and optimization of energy- and area-efficient CMOS synapse and neuron circuits specifically for spiking neural networks that utilize STDP learning capabilities. The authors conduct a thorough analysis of the performance metrics, focusing on minimizing area and power usage while maximizing functionality and adaptability. Detailed experimental validation demonstrates the effectiveness of the proposed circuits in various real-world applications, highlighting their potential to enhance the viability of neuromorphic computing solutions. The findings indicate that such circuits could significantly impact future developments in brain-inspired computing, paving the way for smarter and more efficient processing systems.

2.2.6 Mimicry of Excitatory and Inhibitory Artificial Neurons

[6] J. -K. Han et al., "Mimicry of Excitatory and Inhibitory Artificial Neuron With Leaky Integrate-and-Fire Function by a Single MOSFET," in IEEE Electron Device Letters, vol. 41, no. 2, pp. 208-211, Feb. 2020.

This study investigates the design of artificial neurons that mimic both excitatory and inhibitory functions using a single MOSFET configured to exhibit leaky integrate-and-fire characteristics. The proposed device simplifies the architecture of neuron models while maintaining essential functional capabilities required for effective spiking neural networks. By presenting a detailed analysis of the circuit behavior and its alignment

with biological neuron properties, the authors highlight the potential of this approach to contribute to the advancement of compact neuromorphic systems. The findings suggest that such simplified neuron designs could facilitate more efficient implementations in future SNN architectures.

2.2.7 Integrated Circuits for Energy-Efficient Neurons and Synapses

[7] J. M. Cruz-Albrecht et al., "Energy-Efficient Neuron, Synapse and STDP Integrated Circuits," in IEEE Transactions on Biomedical Circuits and Systems, vol. 6, no. 3, pp. 246-256, Jun. 2012.

The authors introduce integrated circuits designed for energy-efficient neurons, synapses, and spike-timing-dependent plasticity (STDP) mechanisms, focusing on architectures that minimize power consumption while ensuring robust performance. The paper details the various configurations and how they mimic biological systems, providing insights into the fundamental principles of neuromorphic computing. The experimental results validate the effectiveness of these designs in various applications, particularly in neuromorphic computing and biomedical systems, where energy efficiency is critical. The implications of these integrated circuits suggest a significant advancement in the development of bio-inspired computational hardware, paving the way for future innovations in artificial intelligence.

2.2.8 Unsupervised Learning of Digit Recognition Using STDP

[8] P. U. Diehl and M. Cook, "Unsupervised Learning of Digit Recognition Using Spike-Timing-Dependent Plasticity," in Frontiers in Computational Neuroscience, vol. 9, no. 99, 2015.

This paper explores the capability of spiking neural networks (SNNs) to learn digit recognition through unsupervised methods using spike-timing-dependent plasticity (STDP). The authors present a series of experiments demonstrating how SNNs can effectively recognize temporal patterns in input data without supervision, revealing the underlying principles that drive learning in biological neural systems. Their findings provide significant insights into the effectiveness of STDP in real-world recognition

tasks, suggesting that SNNs can achieve performance comparable to traditional neural networks while offering advantages in terms of energy efficiency and temporal processing. This work contributes to the growing understanding of how unsupervised learning paradigms can be harnessed in computational neuroscience.

2.2.9 Energy-Efficient Neuron, Synapse and STDP Integrated Circuits

[9] J. M. Cruz-Albrecht, M. W. Yung, and N. Srinivasa, "Energy-Efficient Neuron, Synapse and STDP Integrated Circuits," in IEEE Transactions on Biomedical Circuits and Systems, vol. 6, no. 3, pp. 246-256, Jun. 2012.

This paper presents integrated circuits specifically designed for energy-efficient neurons, synapses, and spike-timing-dependent plasticity (STDP) mechanisms. The authors provide a detailed examination of the architectures of these circuits, emphasizing their low power consumption while preserving the necessary functionalities to emulate biological neural systems. Through extensive experimental results, the authors demonstrate the effectiveness of their designs in various applications, particularly in neuromorphic computing and biomedical systems, where energy efficiency is critical. The implications of these integrated circuits suggest a significant advancement in the development of bio-inspired computational hardware, paving the way for future innovations in artificial intelligence.

2.2.10 Survey of Encoding Techniques for SNN Signal Processing

[10] D. Auge et al., "A Survey of Encoding Techniques for Signal Processing in Spiking Neural Networks," in *Neural Processing Letters*, vol. 53, pp. 149-171, 2021.

The authors offer a comprehensive survey of encoding techniques utilized in signal processing for spiking neural networks (SNNs). The paper categorizes various encoding strategies based on their efficiency and effectiveness in representing information, examining methods such as rate coding, temporal coding, and others. By analyzing the trade-offs associated with each encoding technique, the authors aim to guide researchers in selecting the most appropriate methods for specific applications, facilitating advancements in the field of spiking neural networks and their real-world implementations.

2.2.11 Memristor-Based Circuit Design for Multilayer Neural Networks

[11] Z. Yang, X. Wang, and E. G. Friedman, "Memristor-Based Circuit Design for Multilayer Neural Networks," in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 65, pp. 1-10, 2017.

The paper presents a detailed analysis of memristor-based circuit designs for implementing multilayer neural networks. It explores the advantages of using memristors to store synaptic weights and demonstrates how memristive crossbar arrays can be leveraged to enhance computation efficiency. The authors highlight the potential of integrating memristors with CMOS circuits to create hybrid architectures that improve scalability and reduce power consumption. The study also discusses design challenges, including variability in memristor behavior and trade-offs between performance and reliability, providing valuable insights for the development of energy-efficient neuromorphic computing systems.

2.3 Conclusion

This literature review highlights the significant progress made in the field of Spiking Neural Networks (SNNs) and their hardware implementations. The examined studies reveal innovative approaches in integrated circuit design, emphasizing energy efficiency while maintaining high performance through the use of CMOS technology and STDP learning mechanisms. Additionally, advancements in encoding techniques underscore the importance of effective signal processing in enhancing the capabilities of SNNs.

Collectively, these contributions illustrate the potential of SNNs to advance neuromorphic computing, paving the way for more sophisticated applications in artificial intelligence and beyond. Future research will need to continue exploring these areas to further optimize performance and energy consumption, ensuring the viability of SNNs in real-world applications.

Chapter 3

SPIKING NEURAL NETWORKS

3.1 Introduction to Spiking Neural Networks

Spiking Neural Networks (SNNs) represent the third generation of neural network models, designed to more closely mimic the behavior of biological brains compared to traditional Artificial Neural Networks (ANNs). Unlike ANNs, which utilize continuous values for neuron activation, SNNs communicate through discrete spikes or action potentials. This spike-based communication allows SNNs to process temporal information, making them particularly suitable for tasks that require real-time processing and learning.

3.2 Structure and Functionality

SNNs consist of interconnected neurons that communicate via spikes. Each neuron in an SNN can be modeled using various mathematical frameworks, with the Leaky Integrate and Fire (LIF) model being one of the most commonly used due to its simplicity and effectiveness in simulating neuronal behavior.

3.2.1 Neuron Models

Several neuron models are employed in SNNs, including:

- **Leaky Integrate and Fire (LIF):** This model integrates incoming spikes over time and generates an output spike when the membrane potential exceeds a certain threshold. The membrane potential then resets, simulating the refractory period of biological neurons.
- **Hodgkin-Huxley Model:** A more complex model that captures the dynamics of action potentials in neurons, accounting for various ionic currents and their interactions.
- **Izhikevich Model:** This model combines the biological realism of the Hodgkin-Huxley model with the computational efficiency of the LIF model, allowing for a wide variety of spiking behaviors.
- **FitzHugh-Nagumo Model:** A simplified version of the Hodgkin-Huxley model that captures essential spiking dynamics while being computationally less intensive.

3.2.2 Types of SNNs

SNNs can be categorized into several types based on their architecture and learning mechanisms:

- **Feedforward SNNs:** These networks have a straightforward architecture where information flows in one direction, from input to output, without cycles.
- **Recurrent SNNs:** These networks include feedback connections, allowing for more complex dynamics and the ability to maintain information over time.
- **Convolutional SNNs:** Similar to convolutional neural networks (CNNs), these networks are designed to process spatial hierarchies in data, making them suitable for image processing tasks.
- **STDP-based SNNs:** Spike-Timing-Dependent Plasticity (STDP) is a learning

rule that adjusts the strength of connections based on the timing of spikes. This biologically inspired learning mechanism allows SNNs to learn from temporal patterns in data.

3.3 Advantages of SNNs

SNNs offer several advantages over traditional ANNs:

- **Temporal Processing:** SNNs can naturally process time-varying signals, making them suitable for tasks such as speech recognition and sensory processing.
- **Energy Efficiency:** The event-driven nature of SNNs allows them to operate with lower power consumption, as neurons only consume energy when they spike.
- **Biological Plausibility:** SNNs provide a more accurate representation of how biological brains function, which can lead to insights into neural computation and learning.

3.4 Challenges and Limitations

Despite their advantages, SNNs face several challenges:

- **Complexity of Training:** Training SNNs can be more complex than training traditional ANNs, particularly due to the discrete nature of spikes and the need for specialized learning rules like STDP.
- **Limited Hardware Support:** While there has been progress in developing hardware for SNNs, many existing implementations rely on traditional digital architectures, which may not fully exploit the advantages of spiking computation.
- **Scalability:** As the size of SNNs increases, maintaining efficient communication and processing can become challenging.

3.5 Applications of SNNs

SNNs have a wide range of applications, including:

- **Robotics:** SNNs can be used for real-time sensory processing and decision-making in robotic systems.
- **Pattern Recognition:** The ability of SNNs to learn from temporal patterns makes them suitable for tasks such as image and speech recognition.
- **Neuromorphic Computing:** SNNs are a key component of neuromorphic systems, which aim to replicate the efficiency and functionality of biological brains in hardware.

3.6 Conclusion

Spiking Neural Networks represent a promising direction in the field of artificial intelligence, offering a more biologically inspired approach to computation. With their ability to process temporal information and operate efficiently, SNNs have the potential to revolutionize various applications, from robotics to pattern recognition. However, addressing the challenges associated with training and hardware implementation will be crucial for their widespread adoption and success.

Chapter 4

Overview of Leaky Integrate and Fire Neuron Model

4.1 Introduction

The Leaky Integrate and Fire (LIF) neuron model is a fundamental concept in computational neuroscience, providing a simplified yet effective representation of neuronal behavior. This model captures the essential dynamics of biological neurons, including the integration of incoming signals, the generation of action potentials (spikes), and the inherent properties of neuronal leakage and refractory periods. This chapter explores the LIF model in detail, incorporating insights from recent research, including the hardware implementation of LIF neurons using memristor technology.

4.2 Overview of the LIF Neuron Model

4.2.1 Basic Dynamics

The LIF neuron model can be described mathematically by the following differential equation:

$$\tau \frac{dV(t)}{dt} = -(V(t) - V_{rest}) + I_{in}(t) \quad (4.1)$$

Where:

- $V(t)$ is the membrane potential at time t .
- V_{rest} is the resting potential of the neuron.
- $I_{in}(t)$ is the input current.
- τ is the membrane time constant, which determines how quickly the membrane potential integrates incoming signals.

4.2.2 Integration of Inputs

When a neuron receives spikes from presynaptic neurons, it generates a small change in its membrane potential. The integration process allows the neuron to accumulate these inputs over time. If the membrane potential reaches a predefined threshold V_{th} , the neuron fires an action potential, resetting its membrane potential to a lower value, often referred to as the reset potential V_{reset} .

4.2.3 Leakage

The "leaky" aspect of the model refers to the gradual decay of the membrane potential over time in the absence of input. This is modeled by a leak conductance that allows the membrane potential to return towards the resting potential, simulating the natural decay observed in biological neurons.

4.2.4 Firing and Refractory Period

Upon reaching the firing threshold, the neuron generates a spike, after which it enters a refractory period during which it is less likely to fire again. This period is crucial for preventing excessive firing and allows for the temporal coding of information.

4.3 Hardware Implementation of LIF Neurons

Recent advancements in neuromorphic engineering have led to the development of hardware implementations of the LIF neuron model. The architecture utilizes mem-

ristor crossbar arrays to create synaptic connections and a custom-designed LIF neuron circuit.

4.3.1 Neuron Circuit Design

The proposed LIF neuron circuit includes several key components:

- **Current Integration:** A capacitor C_u is used to integrate the incoming current.
- **Leakage Mechanism:** A transistor (M5) simulates the leakage of the membrane potential.
- **Spike Generation:** A Schmitt trigger detects when the membrane potential exceeds the firing threshold, generating a spike.
- **Reset Mechanism:** After firing, the membrane potential is reset to zero through a dedicated reset section.

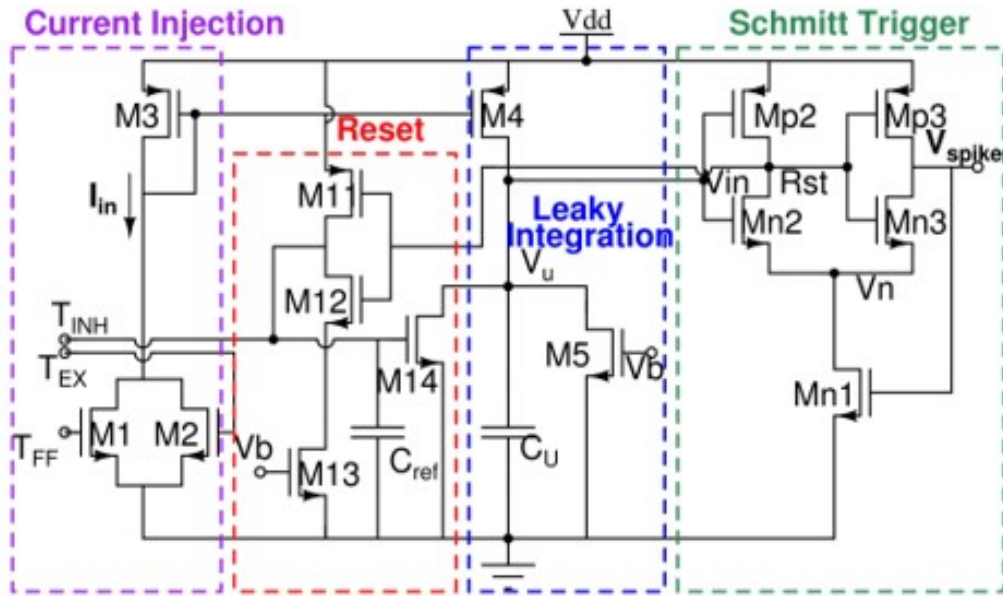


Figure 4.1: Proposed LIF Neuron Circuit

4.3.2 Synaptic Connections

The architecture includes feedforward and lateral excitatory synapses, implemented using memristor crossbar arrays. These arrays allow for adjustable and non-volatile storage of synaptic weights, enabling the system to learn and adapt over time.

4.3.3 Energy Efficiency

The proposed neuron circuit is designed to be energy-efficient, with a static power consumption of approximately 182 pW and energy consumption per spike that can be optimized based on the input voltage. This is crucial for applications in large-scale neural networks where power efficiency is a significant concern.

Parameter	Value
Membrane Time Constant τ	10 ms
Resting Potential V_{rest}	-70 mV
Firing Threshold V_{th}	-50 mV
Reset Potential V_{reset}	-65 mV
Static Power Consumption	182 pW
Energy per Spike	0.4 pJ

Table 4.1: Design Parameters for the Proposed Neuron Circuit

4.4 Conclusion

The Leaky Integrate and Fire neuron model serves as a foundational concept in both theoretical neuroscience and practical implementations in neuromorphic systems. The integration of this model into hardware showcases the potential for creating efficient and adaptive neural networks that mimic biological processes. The use of memristors for synaptic connections further enhances the capability of these systems to learn and store information in a manner similar to biological brains.

Chapter 5

Overview of Pair-based Spike-Timing-Dependent Plasticity (STDP)

5.1 Introduction

Pair-based Spike-Timing-Dependent Plasticity (STDP) is a biological learning mechanism derived from Hebbian theory, which posits that the timing difference between presynaptic and postsynaptic spikes dictates the strength of synaptic connections. In neural networks, STDP is employed to enable temporal learning by adjusting synaptic weights based on the precise timing of spikes. This chapter covers the principles, mechanisms, and implications of pair-based STDP, along with its role in artificial neural network modeling and the critical integration of Leaky Integrate-and-Fire (LIF) neurons in STDP.

5.2 Mechanisms of Pair-based STDP

In pair-based STDP, the synaptic weight modification depends on the temporal interval (Δt) between a presynaptic spike at t_{pre} and a postsynaptic spike at t_{post} . The rule is generally defined as follows:

- **Potentiation** ($\Delta t > 0$): If the presynaptic spike precedes the postsynaptic spike, the synaptic weight is increased (potentiation). This mechanism reflects Hebb's postulate: "cells that fire together, wire together."
- **Depression** ($\Delta t < 0$): If the postsynaptic spike precedes the presynaptic spike, the synaptic weight is decreased (depression), implementing a form of anti-Hebbian learning that helps decorrelate the activity of connected neurons.

The mathematical form of STDP's weight change, Δw , is typically modeled as:

$$\Delta w = \left\{ A_+ \exp\left(-\frac{\Delta t}{\tau_+}\right) \text{ if } \Delta t > 0 - A_- \exp\left(\frac{\Delta t}{\tau_-}\right) \text{ if } \Delta t < 0 \right.$$

where:

- A_+ and A_- are constants defining the magnitude of potentiation and depression, respectively.
- τ_+ and τ_- are time constants controlling the exponential decay rate for potentiation and depression.

5.3 Role of Leaky Integrate-and-Fire (LIF) Neurons in STDP

The Leaky Integrate-and-Fire (LIF) neuron model is fundamental to pair-based STDP, particularly in shaping spike timing and regulating the interactions between presynaptic and postsynaptic spikes. LIF neurons simulate a neuron's spiking behavior by modeling the dynamics of membrane potential over time, which is governed by the differential equation:

$$\tau_m \frac{dV(t)}{dt} = -(V(t) - V_{rest}) + RI(t)$$

where:

- $V(t)$: Membrane potential at time t
- τ_m : Membrane time constant
- V_{rest} : Resting membrane potential

- R : Membrane resistance
- $I(t)$: Input current, representing incoming spike currents

When the membrane potential $V(t)$ reaches a certain threshold V_{thresh} , the neuron fires a spike and resets its potential to V_{reset} . This spiking mechanism, along with the integration of incoming currents, makes LIF neurons ideal for implementing STDP since it provides precise control over spike timing.

5.3.1 Timing Precision in STDP

The LIF neuron model's "leaky" behavior introduces a realistic decay in membrane potential, simulating the natural forgetfulness of biological neurons. This decay enables:

1. **Temporal Dynamics:** The LIF neuron's membrane potential naturally decays when not receiving input, allowing it to capture temporal patterns.
2. **Threshold-based Firing:** Selective firing occurs only when accumulated inputs exceed the threshold, reinforcing temporal learning through STDP.
3. **Spike Synchronization:** By integrating multiple inputs, LIF neurons contribute to aligning spike timings, central to the pair-based STDP learning rule.

5.4 Implementation in Analog STDP Circuits

In analog spiking neural circuits, pair-based STDP with LIF neurons is crucial for creating dynamic and adaptive synaptic connections. Figure 5.1 shows a typical analog implementation of a pair-based STDP circuit, demonstrating how presynaptic and postsynaptic spikes adjust synaptic weights. Analog voltage decay circuits represent the exponential decay component in weight adjustments.

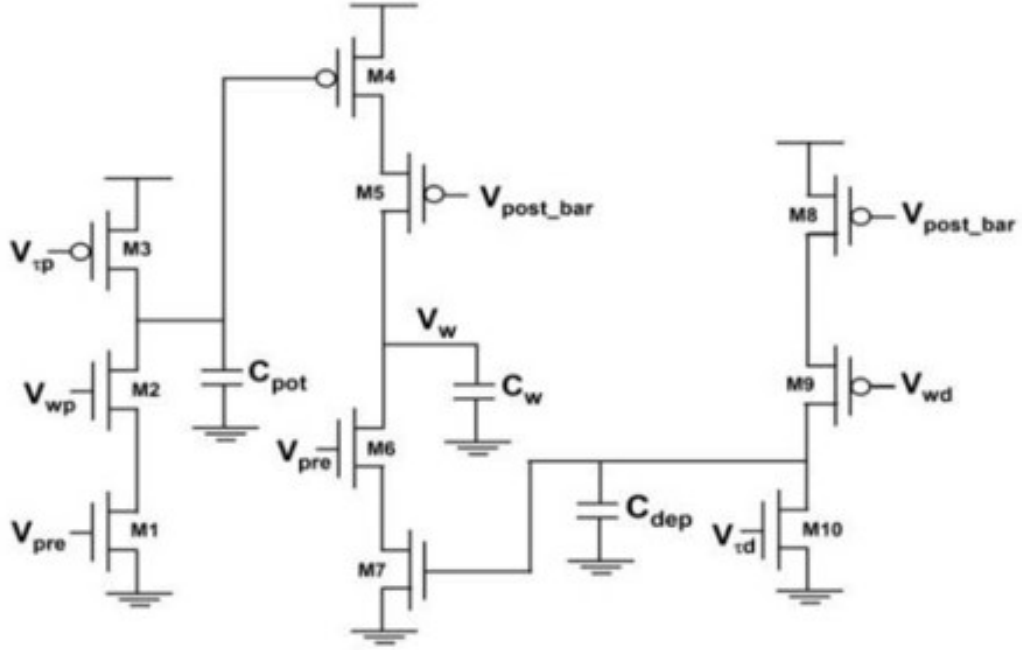


Figure 5.1: Simplified Analog Circuit for Pair-based STDP Implementation

Parameter	Description	Typical Value
A_+	Potential scaling factor	0.005
A_-	Depression scaling factor	0.005
τ_+	Time constant for potentiation	20 ms
τ_-	Time constant for depression	20 ms

Table 5.1: Parameters in Pair-based STDP Circuits

5.5 Mathematical Modeling of STDP

The continuous weight adjustment in pair-based STDP can be expressed as:

$$\Delta w = \alpha e^{-\Delta t / \tau}$$

where α is the learning rate, capturing both the timing relationship and magnitude of changes for efficient adaptation.

Figure 5.2 illustrates the weight change curve for causal and anti-causal pairings, showing exponential decay in weight change as spike timing difference increases, reflecting the biological principle that closely timed spikes have a stronger impact.

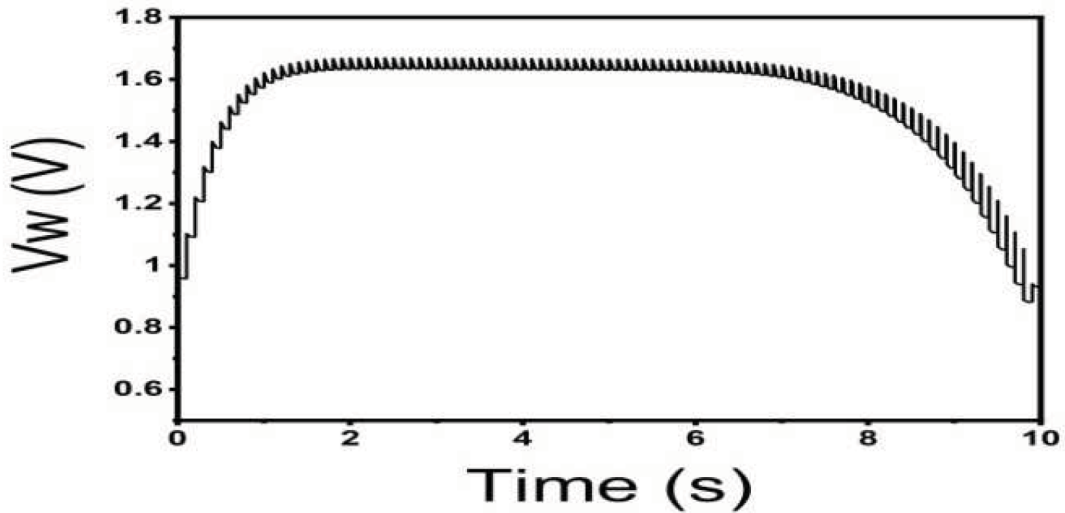


Figure 5.2: Weight Change as a Function of Spike Timing Difference in Pair-based STDP

5.6 Applications of LIF-based STDP Models

Combining LIF neurons with STDP has numerous applications, particularly in tasks requiring temporal learning and pattern recognition:

- **Sensory Processing:** LIF-STDP circuits are beneficial for recognizing patterns in sensory signals, especially in auditory and visual processing.
- **Event-based Sensing:** In event-driven systems, LIF-STDP circuits enable real-time adaptation to stimuli, beneficial in robotics and autonomous systems.
- **Neuromorphic Computing:** LIF-STDP circuits contribute to energy-efficient neuromorphic processors, supporting low-power real-time learning.

5.7 Challenges and Future Directions

Despite its advantages, pair-based STDP with LIF neurons presents challenges, including stability issues due to unbounded synaptic growth or decay, and limitations in capturing complex dependencies beyond pairwise interactions. Future work includes:

- **Hybrid Learning Rules:** Combining pair-based STDP with higher-order STDP rules can improve stability and model complex dependencies.

- **Advancements in Neuromorphic Hardware:** Developing circuits capable of implementing complex STDP with low power consumption.

5.8 Conclusion

The integration of LIF neurons in pair-based STDP enables precise temporal learning, stability, and efficient synaptic adaptation. The combination of the LIF model's threshold-based dynamics with STDP's timing-based weight adjustments allows networks to learn robust temporal patterns, enhancing both biological plausibility and hardware implementations in neuromorphic computing. This chapter has outlined the fundamental principles, modeling techniques, and applications of pair-based STDP with LIF neurons, emphasizing its value in event-driven and adaptive learning tasks.

Chapter 6

Crossbar Array

6.1 Introduction

CMOS-Based Spike-Timing-Dependent Plasticity (STDP) crossbar arrays provide an efficient hardware platform for neuromorphic computing by dynamically adjusting synaptic weights based on spike-timing differences between pre- and postsynaptic neurons. This approach enhances learning capabilities in artificial neural networks, replicating the synaptic plasticity observed in biological systems. The CMOS-based STDP synapses, when integrated with Leaky Integrate-and-Fire (LIF) neurons, enable real-time learning and adaptation, making them suitable for large-scale, energy-efficient neural systems.

6.2 Overview of Crossbar Arrays

Crossbar arrays are fundamental hardware structures used in neuromorphic computing for performing matrix-vector multiplication efficiently. A crossbar array consists of horizontal and vertical metal lines that intersect at certain points, forming a grid where programmable elements, such as memristors or CMOS transistors, are located. These programmable elements control the connection strength (or synaptic weight) between the input and output lines.

The conductance of each element represents the weight of the synapse, allowing

matrix-vector multiplication to be performed in a highly parallel and energy-efficient manner. Crossbar arrays offer a promising approach for implementing artificial neural networks due to their high integration density, non-volatile storage, and suitability for parallel computation.

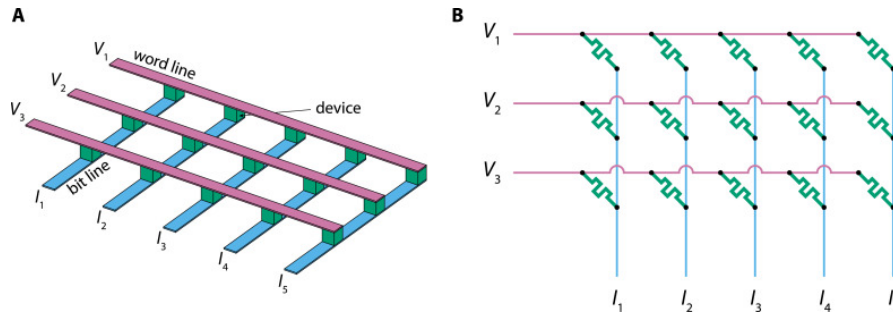


Figure 6.1: Basic structure of a crossbar array used in neuromorphic computing.

6.2.1 Memristive Crossbar Arrays

Memristive crossbar arrays have been widely explored as a hardware solution for synaptic weight storage due to their high density and non-volatility. These arrays utilize memristors arranged in a grid structure where the conductance of individual memristors corresponds to synaptic weights. Weight updates can be achieved using either ex situ or in situ training mechanisms, making memristive arrays a versatile option for neuromorphic applications.

According to Zhang et al. (2017), memristor-based crossbar arrays excel in implementing multilayer neural networks due to their ability to efficiently store and process information in parallel. However, these arrays face significant challenges such as:

- **High Variability:** Memristors suffer from variability in resistance states, which can impact the accuracy of weight updates.
- **Limited Endurance:** Memristors may degrade after extensive programming cycles, leading to unreliable synaptic updates.
- **Error Accumulation:** Stochastic variations can result in cumulative errors, reducing overall system performance.

6.2.2 STDP-Based CMOS Crossbar Arrays

In contrast, CMOS-based STDP crossbar arrays leverage the timing relationship between pre- and postsynaptic spikes to update synaptic weights in real time. These arrays offer several advantages over conventional memristive arrays, including better error tolerance, higher adaptability to real-time scenarios, and improved power efficiency through event-driven weight updates. CMOS circuits provide precise control over spike timing, minimizing variability and ensuring stability in synaptic weight adjustments.

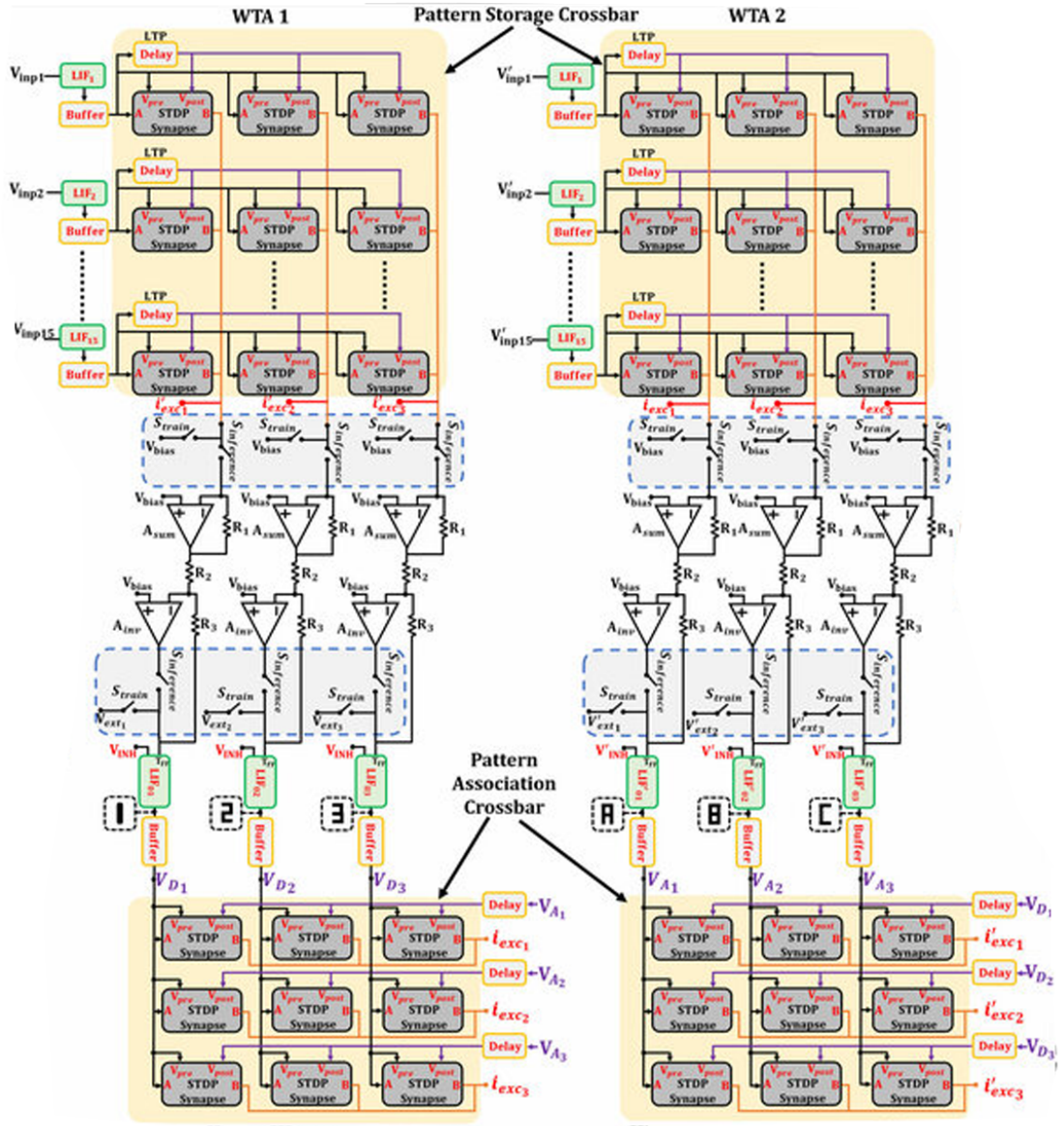


Figure 6.2: Basic structure of a crossbar array used in neuromorphic computing.

6.2.3 Comparison Between Memristive and STDP Crossbar Arrays

The following table provides a detailed comparison between memristive crossbar arrays and CMOS-based STDP crossbar arrays, highlighting their respective advantages and limitations.

Feature	Memristive Crossbar Array	STDP Crossbar Array
Learning Mechanism	External or in situ training	STDP with real-time spike-timing learning
Training Approach	Offline/Online training	Dynamic in situ learning
Synaptic Plasticity	Conductance-based weight updates	Time-dependent STDP weight adjustments
Power Efficiency	Moderate power efficiency	High efficiency due to event-driven updates
Real-Time Adaptation	Limited adaptation capability	High adaptability for real-time scenarios
Hardware Complexity	Lower complexity	Higher due to CMOS-based spike-timing
Error Tolerance	Susceptible to noise and variability	Better tolerance with STDP control
Endurance and Reliability	Limited due to degradation over time	Higher due to stable CMOS fabrication
Scalability and Flexibility	High scalability	Scalability with added control logic

Table 6.1: Comparison Between Memristive Crossbar Arrays and CMOS-Based STDP Crossbar Arrays.

6.3 Advantages of Using CMOS-Based STDP Crossbar Arrays

CMOS-based STDP crossbar arrays outperform conventional memristive arrays by dynamically modifying synaptic weights using a pair-based STDP mechanism. Key advantages include:

- **Real-time Adaptation:** Higher adaptability for real-time learning scenarios.
- **Power Efficiency:** Event-driven updates lead to better energy efficiency.
- **Error Tolerance:** Improved error resilience due to the precise control of spike timing.
- **Stability and Reliability:** CMOS-based circuits mitigate the impact of process variations and degradation.
- **Scalability:** Better suited for large-scale integration with minimal interference.

6.4 Applications of CMOS-Based STDP Arrays

The integration of CMOS-based STDP arrays with neuromorphic architectures has broad applications in artificial intelligence and machine learning:

- **Pattern Recognition:** Efficient for temporal pattern recognition in sensory systems.
- **Neuromorphic Computing:** Ideal for developing low-power neuromorphic processors.
- **Event-Based Sensing:** Enhances performance in event-driven sensing applications such as robotics and IoT.
- **Autonomous Systems:** Enables real-time decision-making in autonomous vehicles and robotics.
- **Speech and Vision Processing:** Suitable for real-time learning and adaptation in

audio and visual systems.

6.5 Challenges and Future Prospects

Despite their advantages, CMOS-based STDP crossbar arrays face challenges such as:

- **Scalability Limitations:** Managing crossbar size while maintaining low power.
- **Process Variations:** Mitigating variations in CMOS fabrication.
- **Stability and Convergence:** Ensuring stable and bounded synaptic weight changes.
- **Hybrid Approaches:** Exploring hybrid CMOS-memristor solutions to overcome the limitations of both technologies.

6.6 Conclusion

CMOS-based STDP crossbar arrays provide an effective platform for implementing real-time synaptic plasticity, improving energy efficiency, and enhancing adaptability in neuromorphic systems. Their advantages over conventional memristive arrays, including better error tolerance and scalability, make them highly suitable for large-scale applications in artificial neural networks and event-driven learning systems. As highlighted by Zhang et al. (2017), integrating memristor-based designs with CMOS-based control mechanisms can further enhance the performance and reliability of crossbar architectures.

Chapter 7

Working Mechanism

7.1 Introduction

The operation of a crossbar array in neuromorphic computing involves processing input spikes, performing efficient matrix-vector multiplication, and generating output spikes that drive subsequent neural layers. CMOS-based Spike-Timing-Dependent Plasticity (STDP) crossbar arrays utilize the precise timing of spikes to dynamically adjust synaptic weights, thereby enabling real-time learning and adaptation. This process mimics biological neural behavior, making it suitable for large-scale, energy-efficient artificial neural networks.

7.2 Working Mechanism of Crossbar Arrays

A crossbar array operates by receiving input signals, processing them through a network of synaptic connections, and generating output responses based on the computed matrix-vector multiplication. This mechanism allows for parallel and efficient neuromorphic computation.

7.2.1 Input Application

In a neuromorphic system, input signals are encoded as spike trains and applied to the crossbar array through the presynaptic neurons. The inputs are typically represented as

voltage pulses, which are fed into the horizontal word lines of the crossbar structure. Each input line corresponds to a neuron, and the strength of the signal is determined by the input spike rate.

Below figure shows a 2x1 pixel patter and it's three possible input classes.

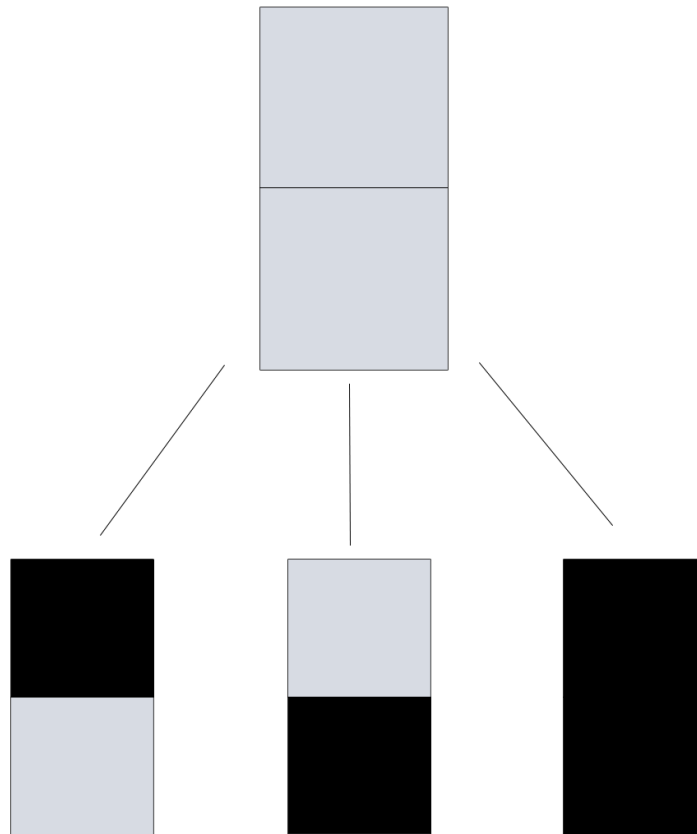


Figure 7.1: Input application in a crossbar array.

7.2.2 Processing in the Crossbar Array

The crossbar array performs matrix-vector multiplication by leveraging the conductance of the synaptic elements. The key steps in the processing mechanism include:

- **Conductance-Based Weighting:** Each intersection in the crossbar contains a programmable synaptic element whose conductance represents the synaptic weight.
- **Current Summation:** When input spikes are applied, the synapses convert them into current based on their conductance values. The total current at each column

(bit line) represents the weighted sum of the inputs.

- **STDP-Based Weight Updates:** The crossbar array dynamically adjusts synaptic weights based on the spike-timing differences between pre- and postsynaptic neurons, reinforcing or weakening synaptic connections accordingly.

7.2.3 Output Generation

The summed current in each column is converted into a voltage signal and processed by the postsynaptic neurons. The output neurons use a thresholding mechanism, such as the Leaky Integrate-and-Fire (LIF) model, to generate spike responses based on the accumulated input. The key aspects of output generation include:

- **Voltage Conversion:** The accumulated current in the crossbar column is converted to a voltage using operational amplifiers.
- **Thresholding Mechanism:** The postsynaptic neuron fires an output spike if the voltage exceeds a predefined threshold.
- **Winner-Take-All (WTA) Mechanism:** In competitive networks, a WTA circuit ensures that only the neuron with the highest activation fires, suppressing other neurons.

7.3 Advantages of Crossbar-Based Computation

Crossbar arrays offer several advantages in neuromorphic computing:

- **Parallel Processing:** Enables simultaneous computation across multiple synapses, improving efficiency.
- **Energy Efficiency:** Reduces power consumption by leveraging analog computation.
- **Scalability:** Supports large-scale integration for complex neural networks.
- **In-Memory Computing:** Eliminates the need for frequent memory access, reducing latency.

7.4 Challenges and Future Enhancements

Despite their benefits, crossbar arrays face certain challenges:

- **Device Variability:** Variations in synaptic elements can impact computation accuracy.
- **Leakage and Noise:** Parasitic currents and signal degradation affect reliability.
- **Scalability Constraints:** Larger arrays require improved fabrication techniques.
- **Hybrid Integration:** Future developments may explore hybrid CMOS-memristor solutions to enhance performance.

7.5 Conclusion

The working mechanism of a crossbar array enables efficient neuromorphic computing by performing matrix-vector multiplication in a highly parallel manner. By leveraging CMOS-based STDP learning, these arrays dynamically adjust synaptic weights, enhancing adaptability and learning capabilities. Continued advancements in crossbar array design will further improve the scalability, accuracy, and energy efficiency of neuromorphic systems.

Chapter 8

Modes of Operation

The crossbar-based neuromorphic system operates in two primary modes: **Training Phase** and **Inference Phase**. These phases play a crucial role in enabling efficient pattern learning and recognition in spiking neural networks (SNNs).

8.1 Training Phase

During the training phase, the synaptic weights in the crossbar array are adjusted dynamically to learn and encode patterns based on spike-timing relationships. The key stages of the training phase include:

- **Input Spike Generation:** Pre-synaptic neurons generate input spikes that propagate through the crossbar array. These spikes are encoded using suitable encoding schemes such as rate coding or temporal coding to represent the information.
- **Weight Update via STDP:** The crossbar array implements a *Spike-Timing Dependent Plasticity (STDP)* mechanism, where synaptic weights are modified based on the relative timing of pre- and post-synaptic spikes. If a pre-synaptic spike precedes the post-synaptic spike, synaptic potentiation occurs, strengthening the connection. Conversely, if the post-synaptic spike occurs before the pre-synaptic spike, synaptic depression reduces the weight.
- **Voltage Application for Learning:** During learning, appropriate control volt-

ages are applied to the word lines and bit lines of the crossbar to enable weight modifications as shown in Fig. 8.1. The applied voltages determine whether potentiation or depression occurs, aligning with the timing and magnitude of spike interactions.

- **Convergence to Stable Weights:** The synaptic weights converge over multiple iterations to achieve an optimized representation of input patterns, ensuring robustness in learned responses.

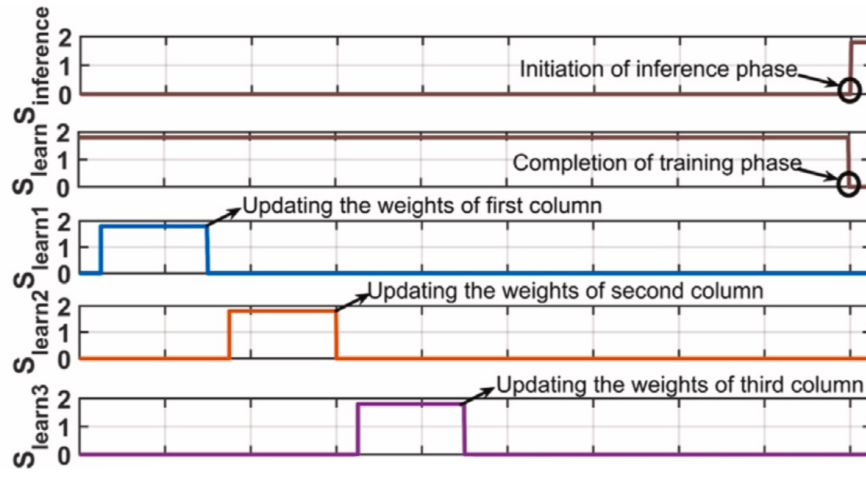


Figure 8.1: Control signals for training each column sequentially

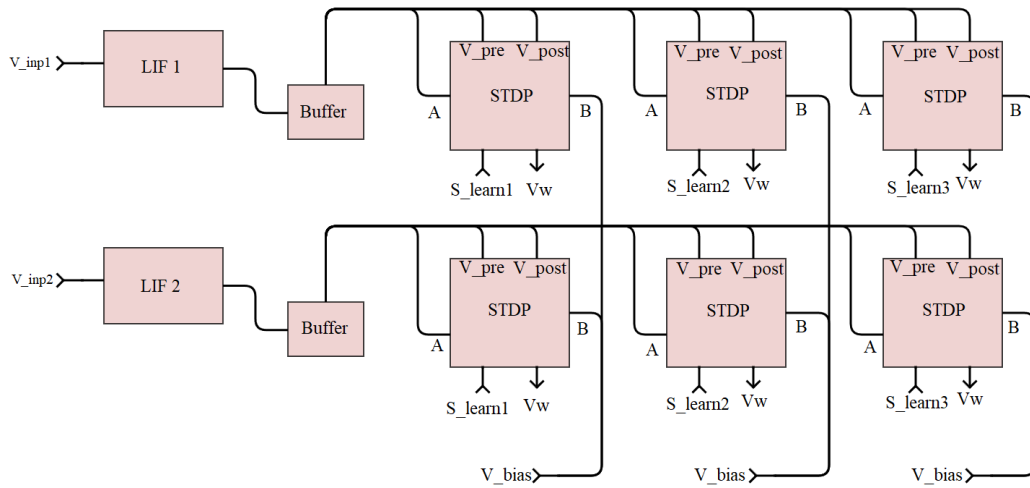


Figure 8.2: Schematic of Training Phase

8.2 Inference Phase

In the inference phase, the trained crossbar array performs real-time pattern recognition by processing new input patterns and generating corresponding outputs. The inference phase involves the following steps:

- **Input Spike Propagation:** Similar to the training phase, pre-synaptic neurons generate input spikes that propagate through the crossbar array. The input is mapped onto the synaptic connections, where the learned synaptic weights influence the response.
- **Summation and Integration:** Post-synaptic neurons integrate the incoming weighted spike signals. The accumulated membrane potential is compared against a threshold to determine the firing of a spike.
- **Output Generation:** When the membrane potential exceeds the threshold, the neuron generates an output spike. This output represents the system's classification or recognition decision.
- **Low-Power Operation:** During inference, the crossbar operates in a low-power mode since synaptic updates are not required, making it highly energy-efficient for real-time applications.

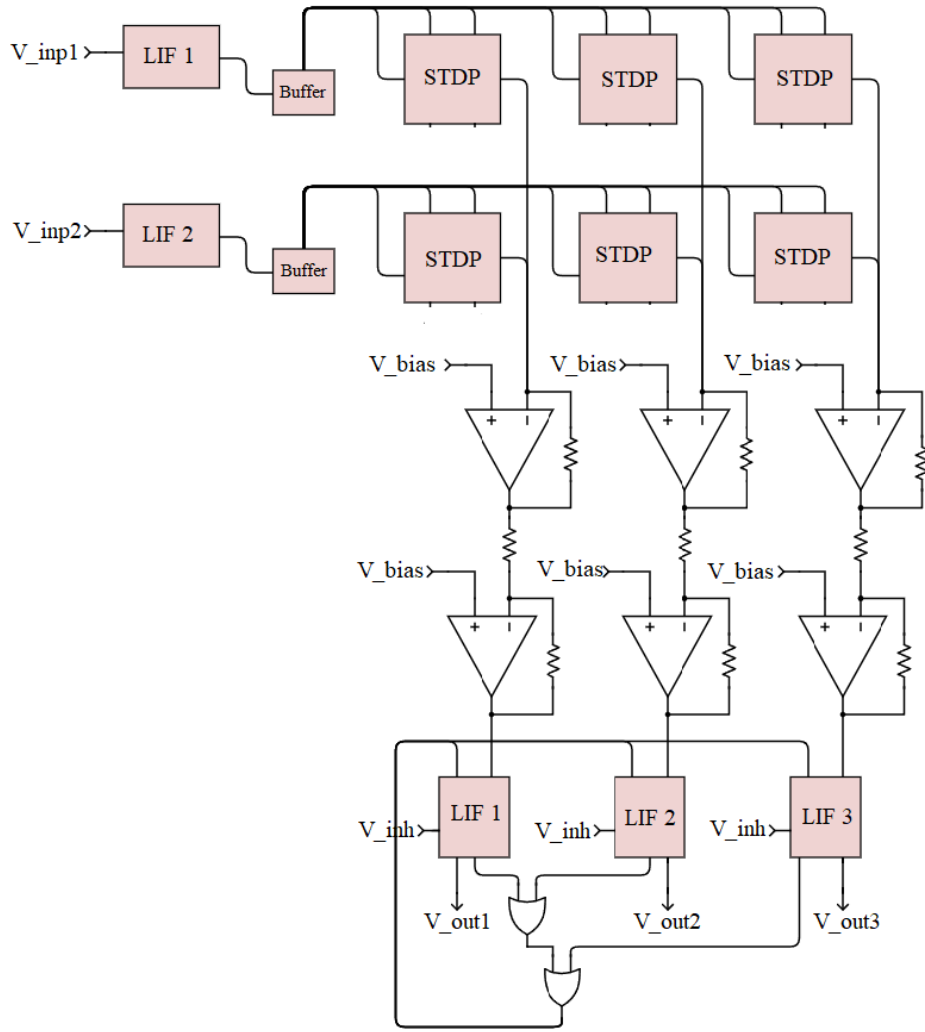


Figure 8.3: Schematic of Inference Phase

8.3 Circuit Modifications for Mode Switching

To switch between the training and inference phases, specific changes must be applied to the circuit configuration:

- Control Signal Configuration:** A mode selection signal is introduced to toggle between training and inference modes. This signal controls the application of appropriate voltages to the crossbar array.
- Training Mode Setup:** In the training mode, higher control voltages are applied to the word lines and bit lines to enable synaptic weight modification. This configuration activates the STDP learning mechanism, allowing weight updates

based on spike timing.

- **Inference Mode Setup:** During inference, the control signal switches to a low-voltage mode where synaptic weights remain fixed. The weight update circuitry is disabled, ensuring that the crossbar performs only forward propagation without modifying synaptic weights.
- **Voltage Pulse Modulation:** Different voltage pulse patterns are required for potentiation and depression during training. These pulses are disabled or bypassed during inference to conserve energy and prevent unintended weight updates.
- **Switching Transistors:** Transistors or multiplexers are used to dynamically switch between the two modes. These components control the path of the applied voltages and ensure that the appropriate circuits are enabled based on the selected mode.

These circuit modifications ensure seamless switching between training and inference modes, allowing the system to efficiently perform learning and pattern recognition tasks.

Chapter 9

About Cadence Virtuoso

9.1 Cadence Virtuoso for Circuit Design and Simulation

Cadence Virtuoso is a comprehensive electronic design automation (EDA) tool suite that is widely used for analog, digital, and mixed-signal circuit design, layout, and simulation. Known for its advanced capabilities and user-friendly interface, Cadence Virtuoso offers designers a seamless environment to conceptualize, create, and verify complex circuits at various levels, from system to layout. For our project on pattern recognition using Spiking Neural Networks (SNN), specifically for implementing Leaky Integrate-and-Fire (LIF) neurons and Spike-Timing-Dependent Plasticity (STDP) synapses, Cadence Virtuoso played a pivotal role in developing and simulating circuit schematics efficiently.

Cadence Virtuoso's schematic editor provides a powerful platform for designing circuits, with support for an extensive library of components, intuitive drag-and-drop functionality, and real-time error checking. This makes it easier to visualize and verify designs, especially when dealing with the intricate and dynamic architectures often required in neural network circuitry. The tool's nuanced simulations capabilities were particularly beneficial in optimizing the neuron and synapse circuits, allowing us to test and fine-tune parameters such as W/L ratios of transistors, capacitances, and voltage

sources under various conditions.

One of the key advantages of using Cadence Virtuoso is its integrated simulation environment. This provided accurate simulations of our LIF neurons and STDP synapses, helping to predict behavior under biologically plausible electrical conditions. This high level of accuracy was essential for validating the timing and interaction between neuron spikes and synaptic plasticity, both of which are crucial for the SNN's learning process. Moreover, the tool's ability to simulate both steady-state and transient responses enabled us to understand neuron firing rates and synaptic weight adjustments over time, making it ideal for simulating the temporally sensitive computations in pattern recognition.

Cadence Virtuoso also offers seamless integration with layout editors, facilitating the transition from schematic to physical layout without compromising design fidelity. This feature was particularly useful for our project, as it allowed us to visualize the spatial configuration of the neuron and synapse circuits and address potential layout-induced issues, early in the design process.

In summary, Cadence Virtuoso provided a comprehensive suite of features that significantly enhanced our circuit design and simulation workflow. Its advanced schematic generation, simulation accuracy, and efficient layout integration made it an invaluable tool in developing the neuron and synapse circuits for this Spiking Neural Network project. This allowed us to focus on fine-tuning circuit performance, thus contributing to the successful implementation of pattern recognition using biologically inspired neural networks.

Chapter 10

Results and Discussion

10.1 Introduction

The objective of this project was to develop and validate a biologically plausible spiking neural network (SNN) model, with components tailored for pattern recognition applications. In the initial phase, two primary components—the Leaky Integrate-and-Fire (LIF) neuron and the pair-based Spike-Timing-Dependent Plasticity (STDP) synapse were designed and tested using Cadence Virtuoso. The successful integration of these circuits into a cohesive model forms the foundation for creating hardware-optimized SNNs that emulate biological learning mechanisms.

This chapter discusses the simulation results obtained from the schematic designs of the LIF neuron and STDP synapse for pattern recognition. Key observations are discussed alongside their implications for real-time neural computation, accuracy, and potential applications in neuromorphic computing.

10.2 Simulations

The simulated CMOS based SNN architecture, combining 2 input neurons, 2x3 crossbar array and 3 output LIF neurons is shown in Fig. 10.1.

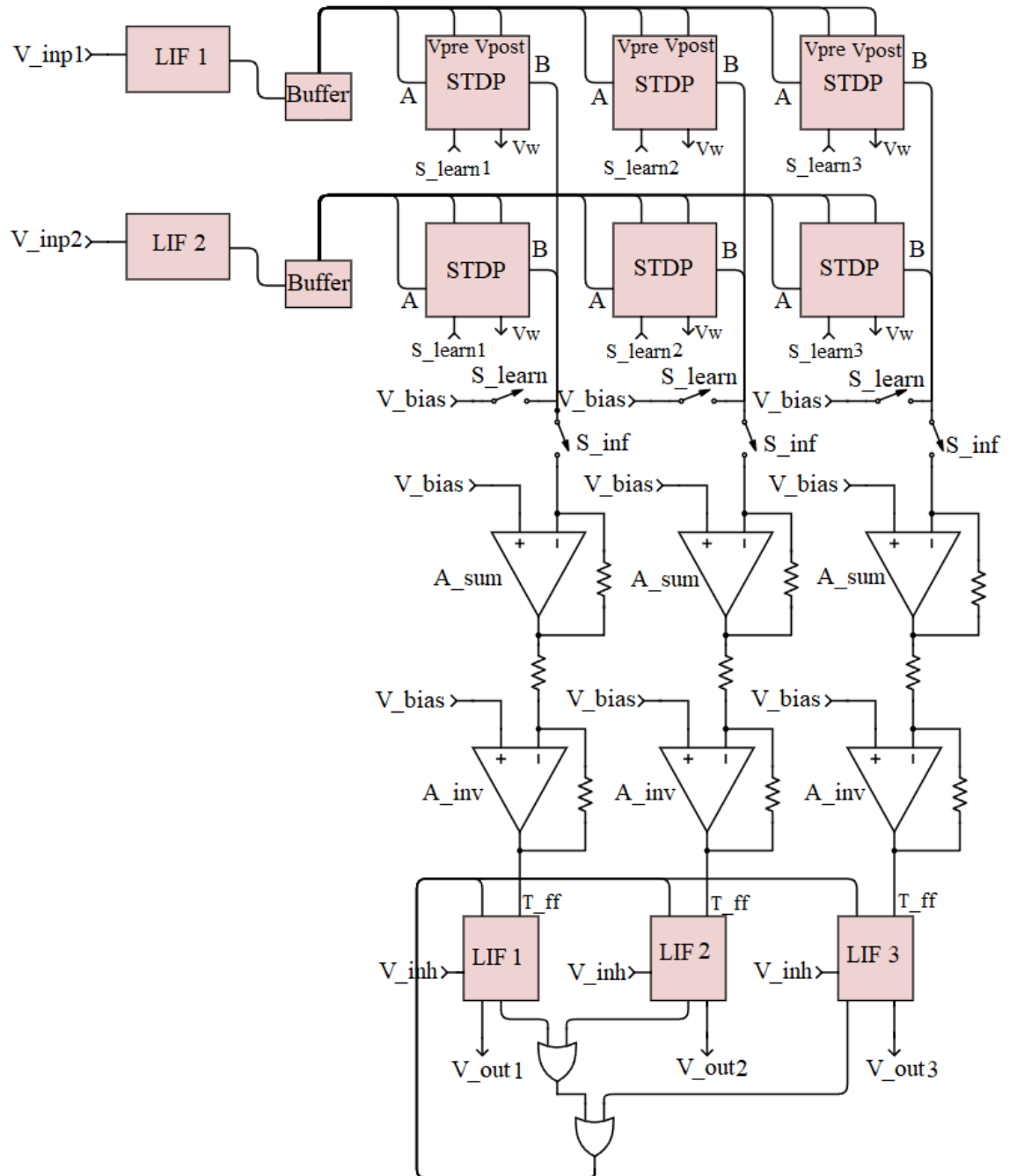


Figure 10.1: Overview of the entire circuit

10.2.1 Leaky Integrate and Fire Neuron

The LIF neuron circuit was designed to emulate biological neuron behavior by firing when the membrane potential crossed a threshold. The design incorporates essential elements, such as capacitive components to model the membrane potential and pathways for the leaky behavior, which discharges the potential over time. Simulation results showed that the circuit accurately replicated expected spiking behavior, validating the neuron's ability to accumulate and release charge in a pattern similar to biological neurons.

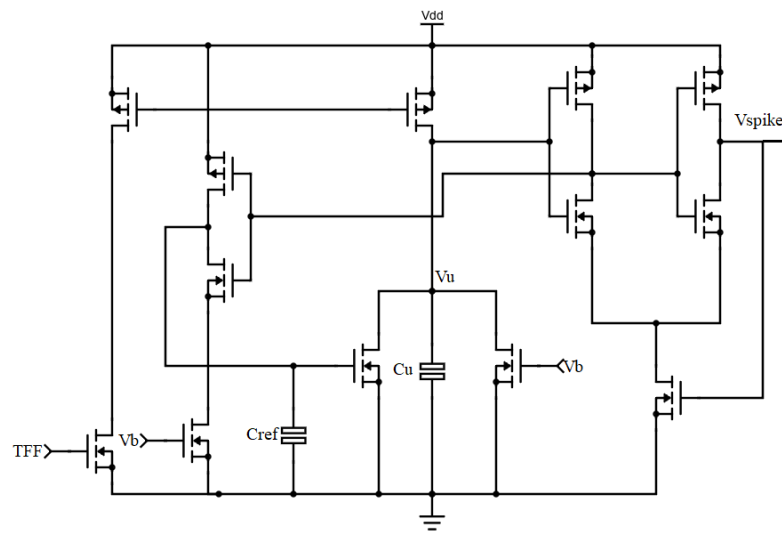


Figure 10.2: Schematic of the Input LIF neuron circuit

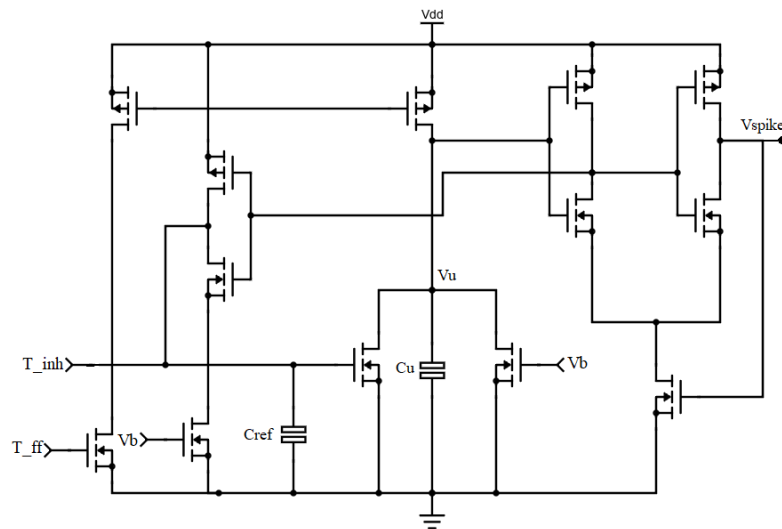


Figure 10.3: Schematic of the Output LIF neuron circuit

The Leaky Integrate-and-Fire (LIF) neuron circuit operates as both the input and output stage of the spiking neural network, with specific circuit modifications for each role. Both input and output circuits are similar except for the external inhibition mechanism in the output LIF. This change in circuit configuration allow the same LIF architecture to efficiently perform different functions while optimizing resource utilization in the neuromorphic system.

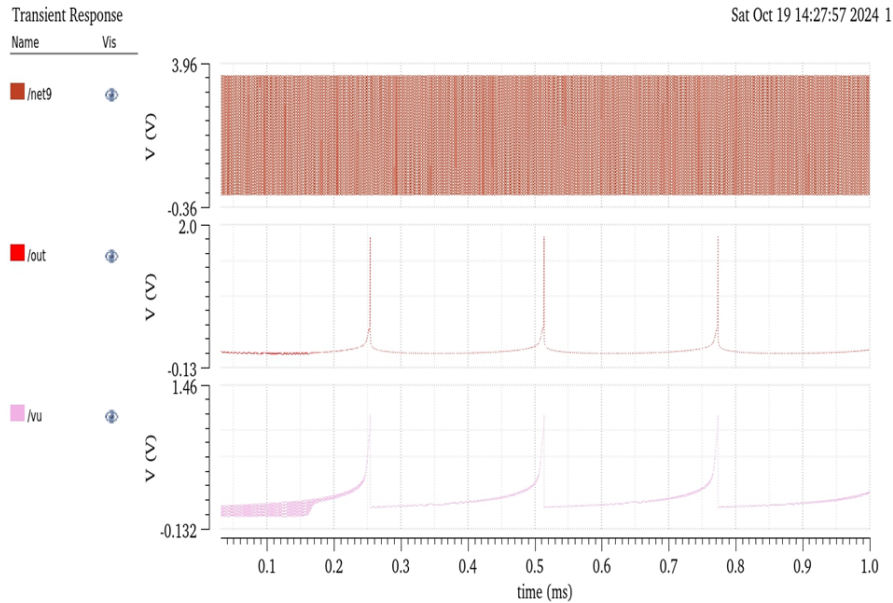


Figure 10.4: Simulation results for the LIF neuron circuit showing membrane potential and spike pattern

10.2.2 Spike Time Dependent Plasticity Synapse

The pair-based STDP synapse schematic was developed to implement synaptic plasticity based on spike timing, where the strength of the synapse increases or decreases depending on the temporal relationship between pre- and post-synaptic spikes. The circuit was successfully simulated, demonstrating the expected weight change behavior consistent with STDP principles. This synapse allows the SNN to adapt its connections dynamically, which is fundamental for learning and memory in neural networks.

The voltage across the synaptic weight capacitor C_w (V_w) is also plotted. It clearly shows the effect of the modified STDP learning rule where the change of the state depends on the rate of the input spike instead of the timing difference. This long-term

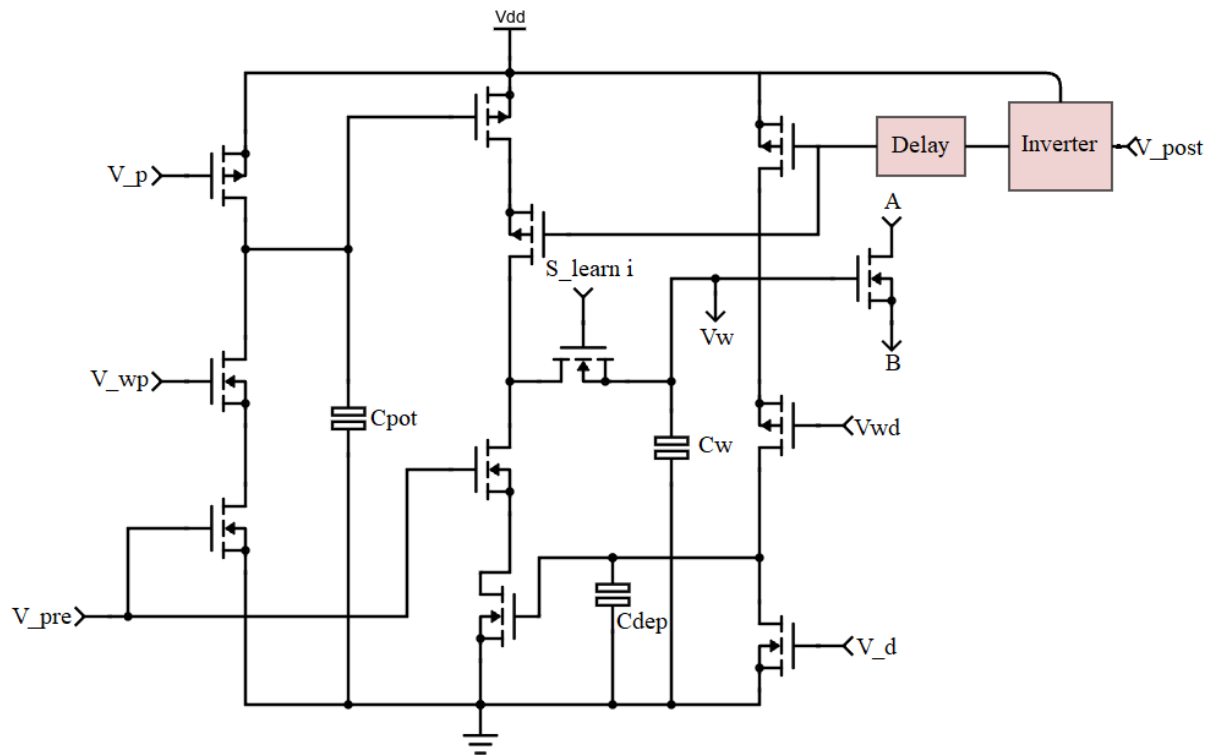


Figure 10.5: Schematic of the STDP synapse circuit

potentiation (LTP) obtained using the memristive STDP circuit is used to update the synaptic weights according to the input.

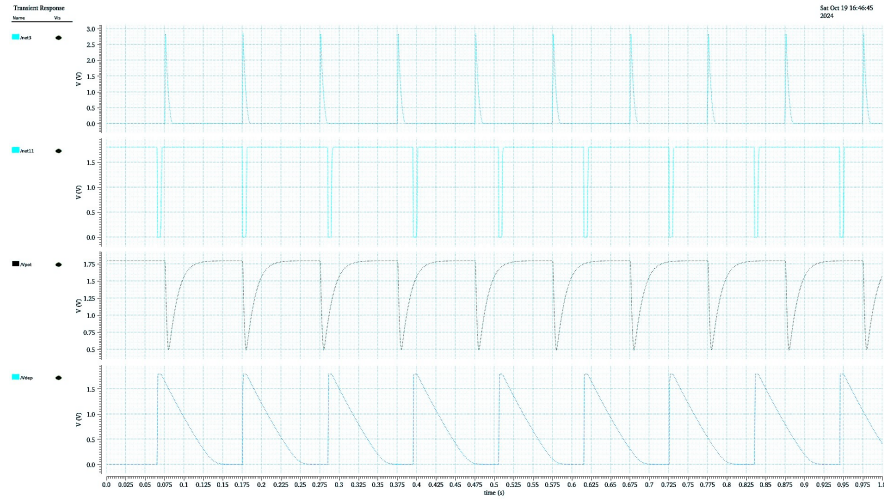


Figure 10.6: Simulation results showing charging and discharging of Vpot and Vdep

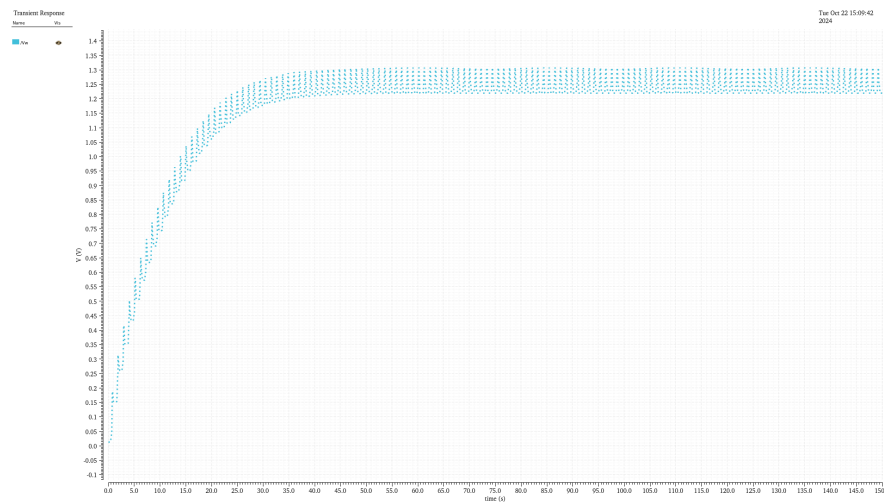


Figure 10.7: Simulation results showing synaptic weight adjustments based on spike timing

10.3 Crossbar Array

The crossbar array architecture is a cornerstone in the design of spiking neural networks (SNN) using memristive synapses. It facilitates the connectivity between the multiple input LIF neurons and output LIF neurons while hosting an array of memristive synapses that leverage spike-timing-dependent plasticity (STDP) for learning.

The proposed crossbar array schematic integrates 2 input LIF neurons with a 6 CMOS memristive synapse circuit array connected to 3 output LIF neurons. Each synaptic connection in the crossbar is designed to adapt the weights according to the

firing timings of the presynaptic and postsynaptic neurons.

The performance of the crossbar array schematic was evaluated through circuit simulations, ensuring effective classification and pattern recognition capabilities of the SNN. The figure given below illustrates the timing dependency between the LIF neurons and STDP synapses.

The output generated from the synapses are a weighted form of the previously generated spikes. The charge stored within the capacitor C_w of each synapse determines its output strength and thus the memristive property is implemented.

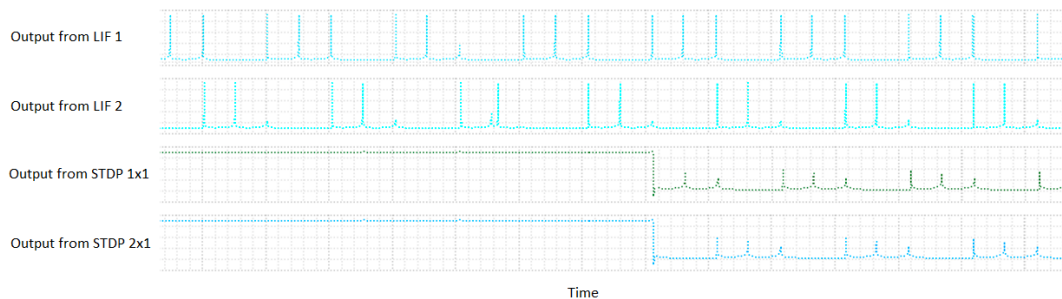


Figure 10.8: Simulation results showing effect of synaptic circuit on the input spikes

After training the network for all patterns, each synaptic crossbar column will store the weights according to the corresponding pattern. Each pattern is presented to the network for a specific amount of time. Only the winner neuron will fire when its corresponding pattern is given at the input while inhibiting the other neurons. Fig. 10.9 shows the firing response of the output LIF neurons, which confirms that each output neuron is capable of recognizing the corresponding pattern.

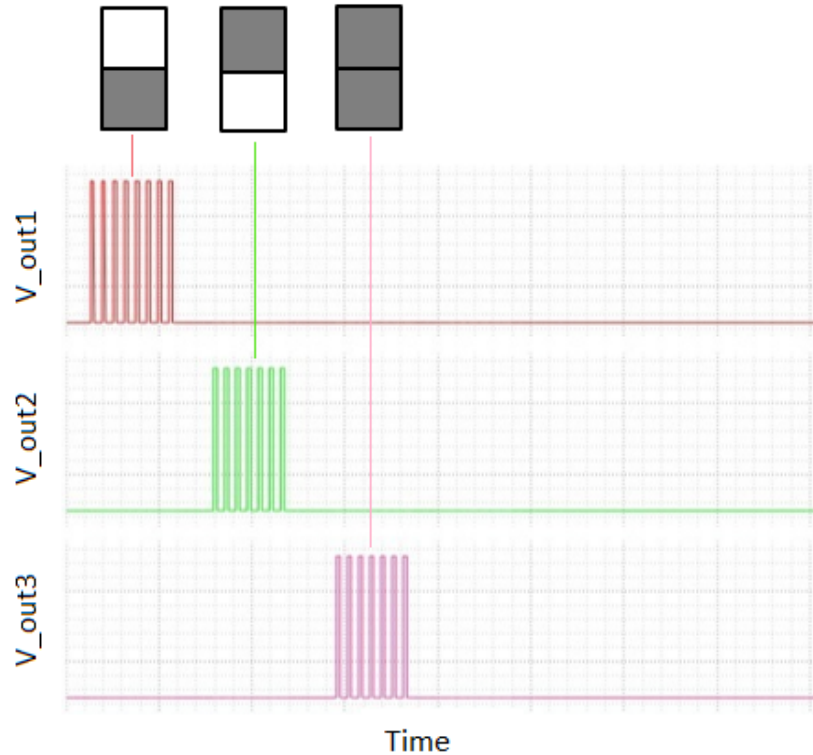


Figure 10.9: Simulation result of classification mode

10.4 Conclusion

Throughout the course of this project, we successfully designed and simulated LIF neuron and pair-based STDP synapse circuits, along with a 2x1 crossbar array, all optimized for pattern recognition. These circuits exhibited accurate electrical behavior that aligns with biological neural networks, enabling effective recognition of various pixel combinations within a 2x1 input matrix. The results suggest that the developed model can be further refined and scaled to address more complex pattern recognition tasks.

Chapter 11

Conclusion

This project focused on the development and implementation of a spiking neural network (SNN) leveraging CMOS technology and memristive synapses. The primary goal was to demonstrate the feasibility of circuit implementations of SNN with a reduced scope compared to the comprehensive architecture detailed in the referenced paper, which proposed a more complex system with a 15x6 crossbar array and 6 output neurons for pattern recognition tasks.

Due to constraints related to computational resources and time efficiency, we successfully realized a simpler 2x3 memristive synapse. This scaled-down model allowed us to explore the fundamental principles of STDP learning and spiking neuron behavior while managing the complexity associated with larger matrix designs. The complexity of the complete circuit proposed in the paper significantly increased the simulation time in Cadence, hampering our ability to perform rapid iterations and adjustments.

The implementation of the 2x1 matrix serves as a proof-of-concept that highlights the potential for developing scalable neural computation systems using CMOS technology and memristive devices. Despite the limited scope, we achieved a foundational understanding of the circuit behavior and synaptic learning characteristics, which provides a basis for future explorations and enhancements.

Future work can build upon this foundation, exploring larger matrix configurations as computational capabilities advance. Additionally, refining the circuit design and

optimizing simulation parameters could enable the realization of more complex and capable SNN architectures while maintaining efficient performance within available technological constraints.

In conclusion, while our project was constrained to a simpler circuit due to available resources, it laid the groundwork for further exploration of neuromorphic computing and CMOS-based SNNs, emphasizing the importance of gradual scaling in the pursuit of more sophisticated neural architectures.

Chapter 12

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Chapter 13

PO Mapping

TITLE	Programme Outcomes (POs)											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Pattern Recognition Using Spiking Neural Networks	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓

PO1 & PO2

- Knowledge of electronic circuits, spiking neural networks, and their applications in pattern recognition was employed.
- Significant efficiency challenges in neuromorphic hardware for pattern recognition were identified and addressed.

PO3

- Developed a compact design integrating CMOS technology with memristive components for improved performance.

PO4

- Conducted investigations into different neuron models and synaptic structures to identify the most suitable for the project.

PO5

- Used Cadence Virtuoso for circuit design and simulation to facilitate the implementation process.

PO6

- Explored real-world applications of spiking neural networks, contributing to advancements in edge computing.

PO8

- Ensured proper citations of all external works and maintained academic integrity in submissions.

PO9

- Worked collaboratively as a team, sharing technical information and coordinating tasks effectively.

PO11

- Conducted detailed planning and scheduling of tasks to ensure project milestones were met within timelines.

PO12

- Presented a thorough literature review identifying gaps and formulating problems to be addressed in the project.